

# Measuring the impact of COVID-19 on cancer survival using an interrupted time series analysis

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## Abstract

**Background:** Few studies have investigated the impact of the COVID-19 pandemic on cancer survival. Those studies that have included pandemic vs prepandemic comparisons can mask differences during different periods of the pandemic such as COVID-19 waves. The objective of this study was to investigate the impact of the COVID-19 pandemic on cancer survival using an interrupted time series analysis and to identify time points during the pandemic when observed survival deviated from expected survival.

**Methods:** A retrospective population-based cohort study that included individuals diagnosed with cancer between January 2015 and September 2021 from Manitoba, Canada, was performed. Interrupted time series analyses with Royston–Parmar models as well as Kaplan–Meier survival estimates and delta restricted mean survival times at 1 year were used to compare survival rates for those diagnosed before and after the pandemic. Analyses were performed for 11 cancer types.

**Results:** Survival at 1 year for most cancer types was not statistically different during the pandemic compared with prepandemic except for individuals aged 50–74 years who were diagnosed with lung cancer from April to June 2021 (delta restricted mean survival times = −31.6 days, 95% confidence interval [CI] = −58.3 to −7.2 days).

**Conclusions:** With the exception of individuals diagnosed with lung cancer, the COVID-19 pandemic did not impact overall 1-year survival in Manitoba. Additional research is needed to examine the impact of the pandemic on long-term cancer survival.

Several studies have investigated the impact of the COVID-19 pandemic on cancer survival (1–5). Most focused on 1 cancer site, and all included prepost comparisons. Unfortunately, the use of simple pandemic vs prepandemic comparisons that provide an average pandemic effect can mask differences in distinct periods of the pandemic such as COVID-19 waves. A prepost study design also does not consider trends in survival that occurred before the pandemic. An interrupted time series analysis (6), however, accounts for baseline trends and helps identify time points during the pandemic period that deviate from expected values. The objective of this study was to examine survival rates at 1 year of follow-up for all individuals diagnosed with cancer over time before and after the start of the COVID-19 pandemic in Manitoba, Canada, using population-based provincial cancer registry data and an interrupted time series analysis.

## Methods

### Data sources

We used data from the Manitoba Cancer Registry, a population-based, high-quality registry that is legally mandated to collect

and maintain accurate, comprehensive information about all individuals diagnosed with cancer in Manitoba (7). The Manitoba Cancer Registry uses International Classification of Diseases for Oncology 3rd Edition disease site groupings. Because stage was determined using the 7th edition of staging by the American Joint Committee on Cancer (AJCC) before 2018 and the 8th edition of staging from 2018 onward, we converted AJCC 8th edition to AJCC 7th edition to adjust for stage in the models for this study.

### Study design

A retrospective cohort study with an interrupted time series analysis was used to examine cancer survival rates prior to COVID-19 (January 2015 to December 2019) and after the start of COVID-19 (April 2020 to September 2021). Individuals diagnosed prior to March 15, 2020, were censored on that date to avoid prepandemic cancer cases having follow-up during the pandemic. Individuals diagnosed from January to March 2020 were excluded because COVID-19 restrictions were implemented throughout March. Quarterly survival rates were analyzed for breast, colon,

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gynecological, head and neck, hematologic, lung, pancreas, prostate, rectal, urinary, and other digestive cancers ([Supplementary Table 1](#), available online, provides International Classification of Diseases for Oncology 3rd Edition codes). Because cancer incidence for some sites decreased in Manitoba during the pandemic (8) and decreases were not consistent across age groups, we divided cancer cohorts into 3 age groups (younger than 50 years, 50-74 years, 75 years and older). Because of low deaths among individuals aged younger than 50 years, results for this age group are only reported for all invasive cancers. Because of high case counts and deaths, subgroup analyses by stage were conducted for lung cancer. Finally, because of high survival for prostate cancer, a subgroup analysis for stage IV cancers was conducted. The end-of-study date was September 30, 2022. The study was approved by the University of Manitoba's Health research ethics board (HS23979; H2020:264).

## Statistical analyses

Details of the methods used for analysis are provided in the [Supplementary Materials](#) (available online). Briefly, Royston-Parmar models (9) were used to analyze survival data, which accounts for time-varying coefficients. Unadjusted and adjusted models were examined. Dummy variables representing different quarters during COVID-19 were considered because changes during the pandemic may correlate with changes in survival that were not accounted for with other time interactions. Dummy variables of 3-month quarter periods were also included to account for seasonality (ie, autocorrelation). A variable indicating time since the beginning of the study period was included to account for baseline trends.

Kaplan-Meier and restricted mean survival time (10) were calculated for fitted (ie, predicted using observed data) and counterfactual estimates (ie, estimates during the COVID-19 period assuming that COVID-19 period had not occurred). Delta restricted mean survival time at 1 year of follow-up between fitted and counterfactual estimates represents the mean survival time lost or gained during the first year of follow-up postdiagnosis during the COVID-19 period; 95% confidence intervals (CI) were produced using bootstrapping.

Forest plots displaying unadjusted and adjusted delta restricted mean survival time estimates, counterfactual estimates, and 95% confidence intervals by cancer site were produced. The 95% confidence intervals of adjusted analyses are highlighted in the text if the interval is within plus or minus 5% or plus or minus 10% of counterfactual estimates, which indicates pandemic survival rates similar to expected values.

Code to run an interrupted time series analysis model with a Royston-Parmar model, simulated data, and R packages used are available in the [Supplementary Materials](#) (available online).

## Results

### All cancer sites

A total of 29 468 individuals in the pre-COVID-19 period (January 2020 to December 2019) and 9236 individuals in the COVID-19 period (April 2020 to September 2021) were included ([Supplementary Tables 2 and 3](#), available online). No statistical differences in unadjusted delta restricted mean survival times were found for all cancers for individuals aged younger than 50 or 50-74 years ([Table 1](#)). Individuals aged 75 years and older diagnosed from April to June 2020 had statistically lower survival (delta restricted mean survival time = -14.1 days, 95% CI = -29.7 to -0.5 days). Those diagnosed from April to June 2021 had statistically higher survival (delta restricted mean survival time = 15.2 days, 95% CI = 2.2 to 28.4 days).

### Lung cancer

Forest plots of delta restricted mean survival times and 95% confidence intervals for lung cancer as well as other cancer sites are presented in [Figures 1 and 2](#) (adjusted analyses) and [Supplementary Figures 1 and 2](#) (available online) (unadjusted analyses). Kaplan-Meier estimates and 95% confidence intervals by cancer site are presented in [Supplementary Figures 3-5](#) (available online). Because of the inclusion of time-varying coefficients in the Royston-Parmar model analyzing lung cancer data, [Supplementary Figure 3](#) (available online) demonstrates 3-month and 1-year Kaplan-Meier estimates. One-year survival for individuals diagnosed with lung cancer from April to June 2021 who were aged 50-74 years statistically decreased by 11% after adjustments for age, sex, and stage

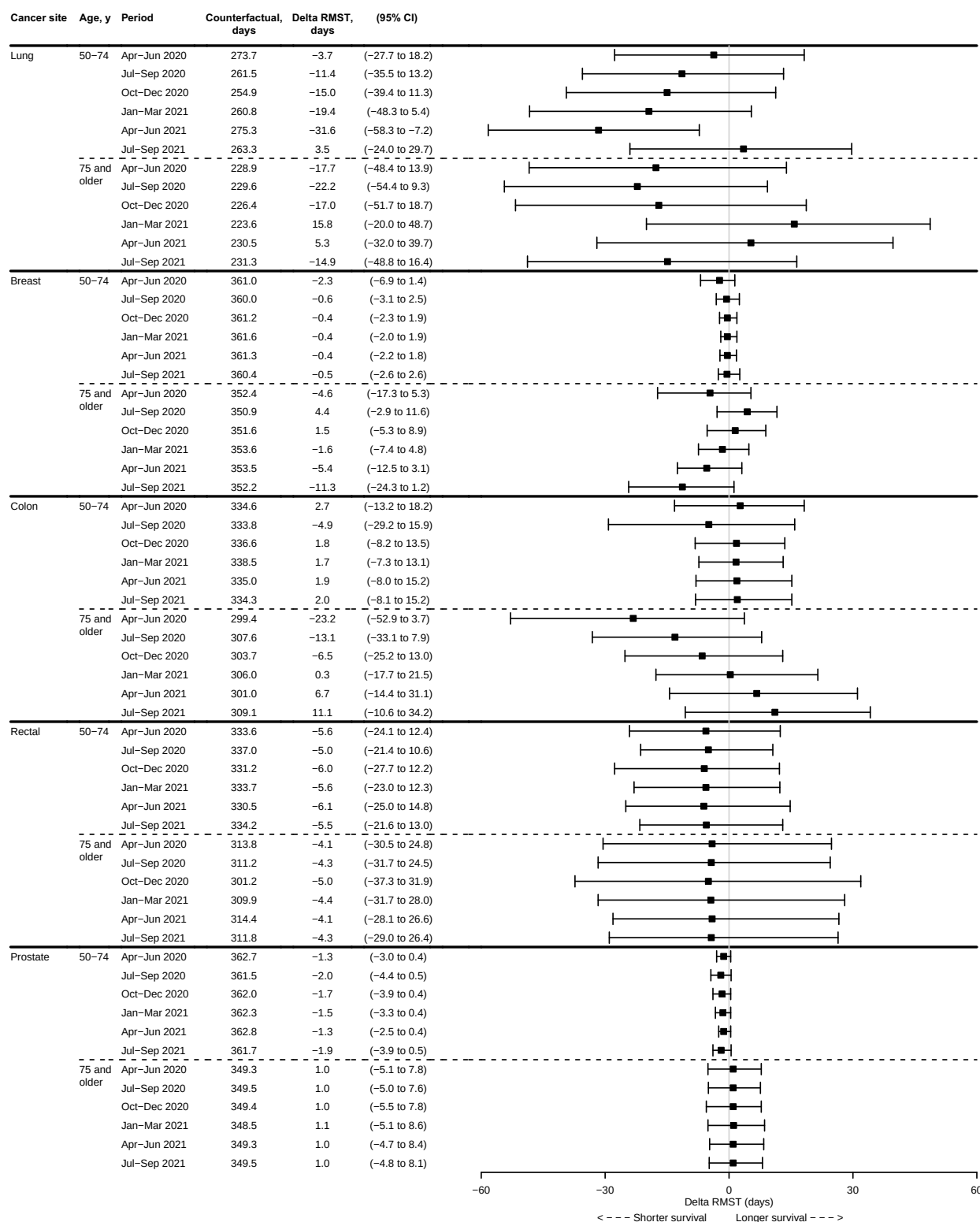
**Table 1.** Unadjusted delta restricted mean survival times and 95% confidence interval for all invasive cancers stratified by age<sup>a</sup>

| Age                   | Period       | No.  | Counterfactual, days | Delta restricted mean survival times (95% CI), days | P     |
|-----------------------|--------------|------|----------------------|-----------------------------------------------------|-------|
| Younger than 50 years | Apr-Jun 2020 | 178  | 345.3                | 5.2 (-2.6 to 13.2)                                  | .2093 |
|                       | Jul-Sep 2020 | 190  | 345.6                | 3.2 (-2.4 to 9.6)                                   | .3006 |
|                       | Oct-Dec 2020 | 202  | 345.4                | 3.2 (-2.3 to 9.8)                                   | .3051 |
|                       | Jan-Mar 2021 | 181  | 347.7                | 2.8 (-2.0 to 8.8)                                   | .3149 |
|                       | Apr-Jun 2021 | 177  | 345.1                | 3.2 (-2.4 to 10.2)                                  | .3205 |
|                       | Jul-Sep 2021 | 172  | 345.3                | 3.2 (-2.3 to 10.3)                                  | .3157 |
| 50-74 years           | Apr-Jun 2020 | 944  | 318.5                | -3.9 (-12.0 to 3.8)                                 | .3322 |
|                       | Jul-Sep 2020 | 992  | 315.1                | -5.5 (-13.5 to 2.6)                                 | .1824 |
|                       | Oct-Dec 2020 | 1090 | 317.7                | -7.5 (-15.3 to 0.0)                                 | .0479 |
|                       | Jan-Mar 2021 | 1063 | 316.1                | 4.4 (-3.1 to 12.7)                                  | .2700 |
|                       | Apr-Jun 2021 | 1093 | 318.8                | -1.7 (-9.5 to 5.7)                                  | .6569 |
|                       | Jul-Sep 2021 | 1071 | 315.5                | -0.8 (-8.2 to 7.2)                                  | .8475 |
| 75 years and older    | Apr-Jun 2020 | 511  | 268.6                | -14.1 (-29.7 to -0.5)                               | .0535 |
|                       | Jul-Sep 2020 | 594  | 267.5                | 9.0 (-3.9 to 21.9)                                  | .1562 |
|                       | Oct-Dec 2020 | 561  | 267.7                | -7.1 (-20.4 to 7.7)                                 | .3053 |
|                       | Jan-Mar 2021 | 528  | 262.7                | 5.7 (-9.0 to 19.3)                                  | .4364 |
|                       | Apr-Jun 2021 | 564  | 268.5                | 15.2 (2.2 to 28.4)                                  | .0240 |
|                       | Jul-Sep 2021 | 534  | 267.5                | -1.2 (-15.9 to 12.2)                                | .8681 |

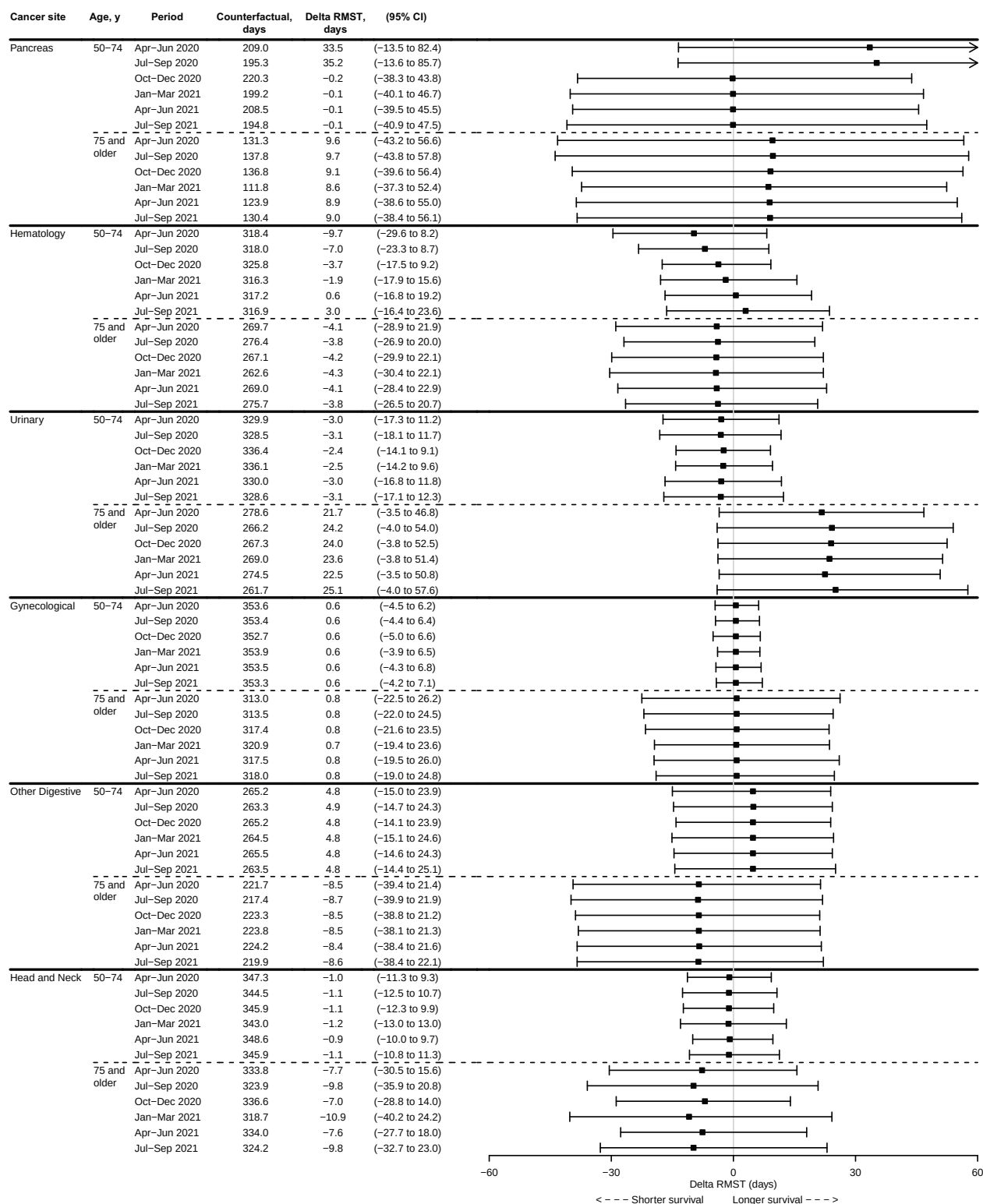
<sup>a</sup> CI = confidence interval.

(delta restricted mean survival time = -31.6 days, 95% CI = -58.3 to -7.2 days). No statistical differences were found for those aged 75 years and older. The first 2 quarters of the pandemic were associated with statistically 5% higher survival for stage I and II lung cancers in adjusted analysis (Table 2). No statistical differences

were found for stage III lung cancer cases. A statistical 11% decrease in survival was found in the second quarter of the pandemic in the adjusted analyses for stage IV lung cancers (delta restricted mean survival time = -27.2 days, 95% CI = -55.3 to -1.4 days).



**Figure 1.** Adjusted restricted mean survival time for lung, breast, colon, rectal, and prostate cancers. CI = confidence interval; RMST = restricted mean survival times.



**Figure 2.** Adjusted restricted mean survival time for pancreatic, hematological, urinary, gynecological, other digestive, and head and neck cancers. CI = confidence interval; RMST = restricted mean survival times.

## Breast cancer

In unadjusted analyses, a statistical decrease in 1-year survival for individuals aged 50-74 years diagnosed with breast cancer was found for the first quarter of the COVID-19 pandemic (delta restricted mean survival time = -9.1 days, 95% CI = -19.2 to -0.04

days). This difference was reduced and became non-statistically different after adjusting for age and stage (delta restricted mean survival time = -2.3 days, 95% CI = -6.9 to 1.4 days). The remaining quarters demonstrated little difference in survival compared with the counterfactual. In addition, 95% confidence intervals

**Table 2.** Unadjusted and adjusted delta restricted mean survival times and 95% confidence intervals for lung cancer (stratified by stage) and prostate cancer (limited to stage IV)<sup>a</sup>

| Site     | Stage | Period       | No. | Unadjusted           |                                      |       | Adjusted             |                                      |       |
|----------|-------|--------------|-----|----------------------|--------------------------------------|-------|----------------------|--------------------------------------|-------|
|          |       |              |     | Counterfactual, days | Delta restricted mean survival times | P     | Counterfactual, days | Delta restricted mean survival times | P     |
|          |       |              |     |                      | (95% CI), days                       |       |                      | (95% CI), days                       |       |
| Lung     | I-II  | Apr-Jun 2020 | 61  | 333                  | 16.1 (3.8 to 29.4)                   | .0134 | 333.9                | 16.1 (3.8 to 28.6)                   | .0089 |
|          |       | Jul-Sep 2020 | 71  | 329.6                | 17.8 (3.8 to 32.5)                   | .0133 | 332.8                | 16.7 (3.9 to 28.9)                   | .0100 |
|          |       | Oct-Dec 2020 | 87  | 330.2                | 7.8 (−6.8 to 21.3)                   | .2806 | 331.1                | 9 (−4.3 to 21.4)                     | .1807 |
|          |       | Jan-Mar 2021 | 71  | 327.4                | 8.4 (−7.4 to 23.3)                   | .2853 | 328.8                | 9.6 (−4.5 to 23.5)                   | .1858 |
|          |       | Apr-Jun 2021 | 82  | 333.3                | 7.1 (−6.1 to 19.8)                   | .2878 | 333.4                | 8.4 (−3.7 to 20.9)                   | .1866 |
|          |       | Jul-Sep 2021 | 63  | 329.9                | −13.6 (−45.3 to 16.2)                | .3975 | 332.3                | −16.4 (−47.7 to 12.9)                | .2846 |
|          | III   | Apr-Jun 2020 | 35  | 271.4                | 5.4 (−42.0 to 53.5)                  | .8282 | 272.4                | −4.6 (−56.5 to 46.1)                 | .8593 |
|          |       | Jul-Sep 2020 | 25  | 269.6                | 12.9 (−40.1 to 63.4)                 | .6207 | 270.4                | 18 (−27.9 to 64.6)                   | .4323 |
|          |       | Oct-Dec 2020 | 35  | 257.4                | 14.6 (−28.1 to 56.3)                 | .4980 | 259.5                | 15.5 (−25.6 to 56.0)                 | .4595 |
|          |       | Jan-Mar 2021 | 37  | 267.2                | 0 (−31.5 to 34.8)                    | .9997 | 266.7                | 1.6 (−30.0 to 38.4)                  | .9290 |
|          |       | Apr-Jun 2021 | 33  | 273.8                | −14.5 (−59.9 to 28.6)                | .5441 | 274.4                | −12.2 (−60.4 to 32.6)                | .6182 |
|          |       | Jul-Sep 2021 | 46  | 272.1                | −38.7 (−82.7 to 5.6)                 | .0912 | 272.4                | −26.1 (−69.3 to 14.3)                | .2254 |
|          | IV    | Apr-Jun 2020 | 118 | 163.1                | 2.6 (−25.0 to 32.0)                  | .8608 | 167.7                | 0.1 (−26.7 to 28.3)                  | .9945 |
|          |       | Jul-Sep 2020 | 132 | 162.4                | −22 (−50.5 to −4.3)                  | .1143 | 166.6                | −27.2 (−55.3 to −1.4)                | .0487 |
|          |       | Oct-Dec 2020 | 114 | 156                  | −22.2 (−49.9 to 8.3)                 | .1377 | 156.2                | −23.8 (−52.7 to 6.3)                 | .1110 |
|          |       | Jan-Mar 2021 | 91  | 147.6                | 3.2 (−32.1 to 37.9)                  | .8553 | 153.6                | −4.3 (−38.7 to 28.0)                 | .7987 |
|          |       | Apr-Jun 2021 | 100 | 164.4                | −12.2 (−44.6 to 21.2)                | .4573 | 170.4                | −18.1 (−50.3 to 14.4)                | .2696 |
|          |       | Jul-Sep 2021 | 104 | 163.8                | 1.0 (−34.0 to 33.7)                  | .9511 | 169.4                | −1.2 (−35.8 to 30.3)                 | .9433 |
| Prostate | IV    | Apr-Jun 2020 | 41  | 327.9                | −0.8 (−15.0 to 15.0)                 | .9206 | 333                  | −3.4 (−15.9 to 10.7)                 | .6155 |
|          |       | Jul-Sep 2020 | 35  | 326.6                | −0.8 (−15.6 to 15.3)                 | .9207 | 331.9                | −3.5 (−16.5 to 11.0)                 | .6175 |
|          |       | Oct-Dec 2020 | 39  | 320.9                | −0.9 (−19.0 to 17.5)                 | .9209 | 329.7                | −3.8 (−18.5 to 11.3)                 | .6192 |
|          |       | Jan-Mar 2021 | 57  | 333                  | −0.7 (−13.3 to 13.8)                 | .9214 | 338.2                | −2.9 (−13.5 to 9.5)                  | .6153 |
|          |       | Apr-Jun 2021 | 47  | 328.1                | −0.8 (−14.0 to 15.9)                 | .9210 | 333.6                | −3.4 (−14.7 to 11.2)                 | .6126 |
|          |       | Jul-Sep 2021 | 47  | 326.9                | −0.8 (−14.7 to 16.3)                 | .9211 | 332.5                | −3.5 (−15.3 to 11.5)                 | .6141 |

<sup>a</sup> CI = confidence interval.

were within plus or minus 5% of the counterfactual estimates for all quarters in the adjusted analyses. For individuals aged 75 years and older, 5 of 6 quarters demonstrated 95% confidence intervals within plus or minus 5% of the counterfactual estimates in the adjusted analyses.

### Colon cancer

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with colon cancer in both age groups. A statistical interaction between the COVID-19 period and time was found ( $P = .017$ ) for the group aged 75 years and older. Of the 6 quarters, 5 demonstrated 95% confidence intervals within plus or minus 5% of the counterfactual estimates for the group aged 50-74 years, and 3 of 6 quarters demonstrated 95% confidence intervals within plus or minus 10% of the counterfactual estimates for the group aged 75 years and older.

### Rectal cancer

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with rectal cancer in both age groups. All quarters for the group aged 50-74 years analyses and 5 of 6 quarters for the group aged 75 years and older demonstrated 95% confidence intervals within plus or minus 10% of the counterfactual estimates.

### Prostate cancer

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with prostate cancer in both age groups. All 95% confidence intervals were within plus or minus 5% of the counterfactual estimates in both age groups. A subgroup analysis that included stage IV prostate cancers demonstrated 95% confidence intervals within plus or minus 5% of counterfactual estimates for 5 of 6 quarters (Table 2).

### Pancreatic cancer

Two quarters from April to September 2020 demonstrated statistically higher survival (delta restricted mean survival time = 48.8 days, 95% CI = 1.9 to 94.0 days; delta restricted mean survival time = 50.2 days, 95% CI = 2.0 to 97.4 days) in the unadjusted analyses for individuals aged 50-74 years diagnosed with pancreatic cancer. However, no statistical differences in survival were found in adjusted analyses for individuals diagnosed with pancreatic cancer in both age groups.

### Hematologic cancers

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with hematologic cancer in both age groups. A non-statistically different interaction between the COVID-19 period and time was included ( $P = .266$ ) for the group aged 50-74 years analysis. All quarters for the group aged 50-74 years demonstrated 95% confidence intervals within plus or minus 10% of counterfactual estimates.

### Urinary cancers

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with urinary cancer for both age groups. Of the 6 quarters, 5 demonstrated 95% confidence intervals within plus or minus 5% of the counterfactual estimates for the group aged 50-74 years.

### Gynecological cancers

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with gynecological cancer for both age groups. In adjusted analyses, all quarters for individuals aged 50-74 years demonstrated 95% confidence intervals within plus or minus 5% of the counterfactual estimates, and all quarters for the group aged 75 years and older

demonstrated 95% confidence intervals within plus or minus 10% of the counterfactual estimates.

### Other digestive cancers

No statistical differences were found in the unadjusted and adjusted analyses for individuals diagnosed with other digestive cancers who were aged 50-74 years. All quarters for the group aged 50-74 years demonstrated 95% confidence intervals within plus or minus 10% of counterfactual estimates. No statistical differences were found for those aged 75 years and older. However, a potential outlier was found in the group aged 75 years and older that is consistent with other observations during the pre-pandemic period. Because of the variability in survival rates between quarters and wide 95% confidence intervals, a main effect model was chosen instead of including a COVID-19-time interaction.

### Head and neck cancers

No statistical differences were found in unadjusted and adjusted analyses for individuals diagnosed with head and neck cancer for both age groups. In the adjusted analyses, all quarters for individuals diagnosed with a head and neck cancer who were aged 50-74 years demonstrated 95% confidence intervals within plus or minus 5% of the counterfactual estimates, and 4 of 6 quarters for individuals aged 75 years and older demonstrated 95% confidence intervals within plus or minus 10% of the counterfactual estimates.

## Discussion

Using an interrupted time series analysis and provincial cancer registry data, we found that the COVID-19 pandemic was not associated with changes in 1-year survival for most cancers in Manitoba, Canada. Most analyses demonstrated 95% confidence intervals within plus or minus 5% or plus or minus 10% of expected values, which suggests that survival was similar for most cancer sites during the pandemic relative to expected survival estimates.

The statistical decrease in survival for individuals aged 50-74 years diagnosed with lung cancer occurred during a pandemic wave (11) that included variants (Alpha and Delta) associated with more severe outcomes (12-14). Individuals diagnosed with lung cancer may have been more vulnerable to COVID-19 infections and complications relative to other cancers. Chronic obstructive pulmonary disease and smoking history are high-risk factors for severe COVID-19 outcomes (15) and are more prevalent among those diagnosed with lung cancer than other cancers (16-18). In contrast, those aged 75 years and older demonstrated non-statistically higher survival in 2021. Our prior research found that lung cancer incidence for this age group was also statistically lower than expected during the same quarters (8). In addition, the COVID-19 infection rate was very high for those aged 80 years and older at the end of 2020 (11). Together, these factors suggest that those with the poorest lung cancer prognosis may have been more vulnerable to COVID-19 infection and may have died prior to their lung cancer being diagnosed, leading to non-statistically higher survival among those who were diagnosed with lung cancer.

Individuals aged 50-74 years diagnosed with breast cancer demonstrated statistically lower survival during the first quarter of the pandemic in unadjusted analyses. This lower survival coincided with the temporary suspension of Manitoba's breast screening program (19) resulting in reduced breast cancer incidence (8). A higher proportion of late-stage cancers diagnosed

during this quarter explain the statistically lower survival and why the survival difference was minimized when stage was included as a control variable. Adjusted analyses for groups aged 50-74 years and 75 years and older demonstrated narrow 95% confidence intervals, which were within plus or minus 5% of expected survival estimates, suggesting survival similar to expected values.

Individuals aged 50-74 years diagnosed with colon cancer during the second quarter of the pandemic had a non-statistical decrease in survival. Similar to breast cancer, this coincides with a decrease in the number of fecal occult blood tests conducted through Manitoba's colorectal screening program (19). Others have also reported lower colonoscopies and flexible sigmoidoscopies rates at the start of the pandemic (20). These factors explain why colon cancer incidence was statistically lower during the first few months of the pandemic (8), which could have increased the stage distribution at diagnosis. However, several studies have demonstrated little to no change in colon cancer stage distribution during the pandemic (4,21-26). Most quarters for the group aged 50-74 years demonstrated 95% confidence intervals within plus or minus 5% of expected values, suggesting that survival during the pandemic was similar to expected values.

Analyses for individuals diagnosed with prostate cancer demonstrated 95% confidence intervals within plus or minus 5% of 1-year counterfactual values suggesting survival similar to expected values. The subgroup analysis that included only stage IV cancers also found that most quarters demonstrated 95% confidence intervals within plus or minus 5% of counterfactual estimates, suggesting survival similar to expected values.

Other studies examining the impact of the pandemic on survival in cancer cohorts have found mixed results. Individuals diagnosed with colorectal, non-Hodgkin lymphoma, or uterine cancer demonstrated worse survival in Alberta, Canada (4). In Ontario, Canada, individuals diagnosed with melanoma demonstrated lower survival, and those diagnosed with hepatobiliary or lung cancer demonstrated higher survival (2). No change in pancreatic cancer survival was observed in the Netherlands (3). Survival in the United Kingdom was lower for pancreatic ductal adenocarcinoma, although the relationship was not statistically different when adjusted for control variables (5). Survival for esophagogastric cancer was lower in Scotland but was not statistically different when adjusted for control variables (1). This highlights the importance of adjusting for stage at diagnosis when examining survival. For example, individuals diagnosed with pancreatic cancer during the first 6 months of the pandemic were diagnosed at a later stage. However, pancreatic cancer incidence was lower during this period, which suggests that the increase in stage was because fewer early stage cases were diagnosed. Not adjusting for cancer stage for this period will exaggerate the impact of the pandemic on survival. This contrasts with survival analyses that include individuals diagnosed in later periods of the pandemic (2021 and later) when delayed diagnoses are captured leading to a higher stage distribution. If a shift in stage has occurred because of delayed diagnoses and cancer incidence has returned to or exceeded expected rates, survival models including stage as a control variable for this period will assume that pre-pandemic and pandemic periods have the same stage distributions eliminating the impact of stage distribution shift. This will underestimate the impact of the pandemic on survival.

The Royston-Parmar model was used to analyze survival data, which allowed time-varying effects to be accounted for, such as those found in lung cancer analyses (ie, larger survival

differences at 3 months than 1 year). Restricted mean survival times were used to describe survival differences during the pandemic. The difference between fitted and counterfactual estimates in number of days were calculated, which is easily interpretable and can be determined when using a survival model that includes time-varying effects (10). Dummy variables were included when appropriate during the pandemic as COVID-19-period interaction terms to account for changes in survival. This was used in the lung cancer analysis where changes in survival were associated with the Alpha and Delta waves of the COVID-19 pandemic. Prior survival studies used prepost comparisons where the entire pandemic period was compared with the prepandemic period, which does not account for variation in associations over time. An interrupted time series analysis was used, which accounts for baseline trends in survival that are ignored with prepost comparisons. We stratified analyses by age, which improved comparisons for sites where cancer incidence varied by age because of lower cancer screening (breast and colorectal cancer) or lower incidence among older individuals (lung cancer). Lastly, our study included individuals diagnosed up to September 30, 2021. Prior survival studies only included cancer diagnoses during 2020, except for 1 (5), which included individuals up to March 31, 2021.

Limited power for some analyses may be present (ie, individuals aged 50-74 years diagnosed with breast or prostate cancer). Pancreatic cancer had few events, which provided unstable survival rates and variations in estimates between quarters. The 95% confidence intervals for breast and prostate cancer estimates for those aged 50-74 years are too narrow because bootstrapped estimates reached the upper boundary of 100% survival at 1 year of follow-up and are therefore less reliable (27,28). One-year survival may be too short to demonstrate an impact from the pandemic, and long-term survival analyses are required. Because of the number of analyses, the statistical differences in lung cancer survival could be random associations. Additional research is recommended to investigate the incidence of COVID-19 infection among individuals diagnosed with lung cancer relative to other cancers. This study did not include cancers diagnosed during the Omicron wave during which more COVID-19 infections were reported than during previous waves (29). Updated analyses will be done to investigate the impact of Omicron on survival and the impact of a shift in stage at diagnosis.

We found no evidence that the COVID-19 pandemic was associated with a statistical decrease in cancer survival for most cancers at 1-year follow-up; 95% confidence intervals were within plus or minus 10% of expected values, which could suggest that survival rates during the pandemic were similar to the rates expected if the pandemic had not occurred. Changes in survival for individuals diagnosed with lung cancer can be explained by increased vulnerability to COVID-19 infection during the Alpha and Delta waves of the pandemic. Research is needed to examine the impact of the pandemic on long-term cancer survival.

## Data availability

The data that support the findings of this study are not publicly available to ensure and maintain the privacy and confidentiality of individuals' health information. Requests for data may be made to the appropriate data stewards (CancerCare Manitoba's Research and Resource Impact Committee).

## Author contributions

Pascal Lambert, MSc (Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology; Writing—original draft; Writing—review & editing), Katie

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## Conflicts of interest

The authors declare no conflict of interest.

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